

## Lab Exercise: Biological Data Classification Using Machine Learning Method

**Objective:** Implement and evaluate a machine learning model to classify biological samples based on a provided or chosen dataset.

*CLO 2 : Implement, optimize, and analyze computational methods using appropriate tools to interpret biological data.*

---

### Exercise Structure (Total: 10 Marks)

#### Part 1: Data Preparation & Exploration (2 Marks)

- **Tasks:** Load your chosen dataset (e.g., gene expression, protein levels, or clinical data) and perform basic cleaning. Identify the target classes and visualize the distribution of labels (e.g., using a bar chart).
- **Deliverable:** A summary of dataset dimensions and one visualization of class distribution.

#### Part 2: Model Implementation (4 Marks)

- **Tasks:** Split the data into training (80%) and testing (20%) sets. Implement a chosen machine learning classifier (e.g., Logistic Regression, Random Forest, or SVM) using Scikit-learn.
- **Deliverable:** Code showing the model training process and predictions on the test set.

#### Part 3: Evaluation & Short Discussion (4 Marks)

- **Tasks:** Calculate the model's **Accuracy** and display a **Confusion Matrix**. Provide a brief 3–5 sentence interpretation of the results, noting any biological significance or model limitations (e.g., overfitting or small sample size).
- **Deliverable:** Evaluation metrics and a concise summary of findings.

Students are encouraged to refer to the following tutorial to support their understanding of machine learning workflow, particularly classification models using Scikit-learn:

[SVM Classification Tutorial \(DataCamp\)](https://www.datacamp.com/tutorial/svm-classification-scikit-learn-python)

(<https://www.datacamp.com/tutorial/svm-classification-scikit-learn-python>)

It covers key concepts such as:

- Data preprocessing and scaling
- Train-test splitting
- Model training using Scikit-learn
- Prediction and evaluation

- Performance metrics interpretation

Adapt the provided workflow to the chosen dataset used in this laboratory exercise  
Modify the Python code to suit the dataset choose

You may replace or compare the algorithm with other machine learning methods such as:

- Logistic Regression (recommended baseline)
- K-Nearest Neighbors (KNN)
- Random Forest Classifier (optional extension)

### Assessment Rubric (10 Marks)

| Criteria                      | Description                                                         | Marks |
|-------------------------------|---------------------------------------------------------------------|-------|
| <b>Data Handling</b>          | Successful data loading, cleaning, and basic visualization.         | 2     |
| <b>ML Implementation</b>      | Correct data splitting and implementation of an ML algorithm.       | 4     |
| <b>Results &amp; Analysis</b> | Accurate performance metrics and logical interpretation of results. | 4     |

---

### Adjusted Time Allocation (Total: 90 Minutes)

Based on the original suggested timings, the following schedule ensures completion within the 1 hour 30 minute limit:

- **20 Minutes:** Data Acquisition & Cleaning (loading and handling missing values).
- **20 Minutes:** Basic Exploratory Data Analysis (visualizing class distributions).
- **30 Minutes:** Model Training & Evaluation (implementing the classifier and generating metrics).
- **20 Minutes:** Interpretation & Final Submission (brief discussion and notebook export).

**Note:** Students are encouraged to use their own FYP dataset or can choose from

- <https://file.biomedcentral.com/1471-2107/1/1>
- <https://www.kaggle.com/discussions/general/30222>
- <https://archive.ics.uci.edu/>

Quick resources:

- <https://www.youtube.com/watch?v=E0Hmnixke2g> (ML Algorithm)

- <https://www.youtube.com/watch?v=Kdsp6soqA7o> (Confusion Matrix)
- <https://www.youtube.com/shorts/QSoEWNJg3B4> (Result)
- <https://www.youtube.com/shorts/As95IkgMBHM> (Random Forest)
-